Physics 362 - Homework 7 Solutions - Delon Shen

Problem 1: The goal of this problem is to get you familiar with the Zeldovich approximation. We'll do things in the Newtonian limit and assume that there is only matter in the universe ($\Omega_{m,0} = 1$).

Consider a test particle embedded in a dark matter distribution with mass density ρ. This particle has a physical location $\boldsymbol{r}$ and velocity $\boldsymbol{u}$. The equations governing the motion of this particle are simply

$$\frac{d\boldsymbol{r}}{dt} = \boldsymbol{u}, \quad \frac{d\boldsymbol{u}}{dt} = -\frac{\partial \Phi}{\partial \boldsymbol{r}}, \quad \frac{\partial}{\partial \boldsymbol{r}} \cdot \left(\frac{\partial \Phi}{\partial \boldsymbol{r}} \right) = 4\pi G \rho.$$

(a) [3pt] Let $\boldsymbol{x} = \boldsymbol{r}/a$ be our usual comoving coordinates. The physical velocity $\boldsymbol{u}$ then has two components when written in terms of $\boldsymbol{x}$: $\boldsymbol{u} = \dot{\boldsymbol{r}} = a\dot{\boldsymbol{x}} + H\boldsymbol{r}$. The second term is due only to the expansion of the universe while the first term we define as the **peculiar velocity** $\boldsymbol{v} = a\dot{\boldsymbol{x}}$. Similarly we can define a **peculiar potential** $\phi = \Phi + \frac{1}{2}a\ddot{a}\boldsymbol{x}^2$. Show in terms of these variables that our evolution equations becomes

$$\frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{p}}{a^2}, \quad \frac{d\boldsymbol{p}}{dt} = -\nabla\phi, \quad \nabla^2\phi = 4\pi G \bar{\rho}_{m,0}\delta/a, \tag{1}$$

where we defined the density contrast δ as $\rho = \bar{\rho}(1+\delta)$, comoving momentum $\boldsymbol{p} = a\boldsymbol{v}$, and let $\nabla = \partial/\partial\boldsymbol{x}$.

Hint: You may need to use the Raychaudhuri equation applied to a matter dominated universe $\ddot{a}/a = -4\pi G\bar{\rho}/3$ which comes from the spatial part of the Einstein equation for a FLRW metric. See e.g. Eq.(2.137) of Baumann's book.

Solution: The first equation, $d\boldsymbol{x}/dt = \boldsymbol{p}/a^2$, is true just from plugging in $\boldsymbol{v}$ into the definition of $\boldsymbol{p}$. Since $\partial/\partial\boldsymbol{r} = (1/a)\partial/\partial\boldsymbol{x}$ we know that

$$\frac{d^2\boldsymbol{r}}{dt^2} = x\ddot{a} + a\ddot{x} + 2\dot{a}\dot{x}a = -\frac{\nabla\Phi}{a}.$$

The term proportional to position causes a spatial dependence on the particle acceleration which arises from expansion of spacetime. So we can fold it into the definition of a peculiar potential $\phi = \Phi + \frac{1}{2}a\ddot{a}\boldsymbol{x}^2$ leading us to the second equation

$$a^2\ddot{\boldsymbol{x}} + 2a\dot{a}\dot{\boldsymbol{x}} = \frac{d(\dot{\boldsymbol{x}}a^2)}{dt} = \frac{d\boldsymbol{p}}{dt} = -\nabla\phi.$$

We can compute $\nabla^2\phi$ as

$$\nabla^2\phi = \nabla^2\Phi + \ddot{a}a = a^2\frac{\partial}{\partial\boldsymbol{r}}\cdot\left(\frac{\partial\Phi}{\partial\boldsymbol{r}}\right) + \ddot{a}a = 4\pi G\rho_{m,0}\delta/a,$$

where used the Raychaudhuri equation for $\ddot{a}$ and the Poisson equation for $\nabla^2_{\boldsymbol{r}}\Phi$

(b) [2pt] The Zeldovich approximation essentially boils down to the ansatz that particles move in straight lines in the direction of their initial displacement:

$$\boldsymbol{x}(t) = \boldsymbol{q} - c(t)\boldsymbol{f}(\boldsymbol{q}). \tag{2}$$

Here $\boldsymbol{q}$ is the initial (Lagrangian) position of the particle and $\boldsymbol{f}(\boldsymbol{q})$ is the direction of the initial displacement determined by the initial conditions at Lagrangian position $\boldsymbol{q}$. We will determine $c(t)$ shortly.

From conservation of mass $\bar{\rho}(1+\delta(\boldsymbol{x},t))d^3\boldsymbol{x} = \bar{\rho}d^3\boldsymbol{q}$, show that the density contrast evolves like

$$1+\delta = \frac{1}{(1-c(t)\lambda_1)(1-c(t)\lambda_2)(1-c(t)\lambda_3)},$$

where λ_i are eigenvalues of the **deformation tensor** $\partial f_i/\partial x_j$.

Solution: From conservation of mass, we can see that $1+\delta$ evolves like the Jacobian of the coordinate transform from $\boldsymbol{x}$ to $\boldsymbol{q}$:

$$1+\delta = \left|\left|\frac{\partial\boldsymbol{x}}{\partial\boldsymbol{q}}\right|\right|^{-1},$$

where $||A|| = \det A = \prod_i \lambda_i$ for eigenvalues λ_i of the matrix A. Using the ansatz Eq.(2) tells us that

$$\frac{\partial x_i}{\partial q_j} = \delta_{ij}^{(K)} - c(t)\frac{\partial f_i}{\partial q_j},$$

where $\delta^{(K)}$ is the Kronecker delta. Writing $\det A$ as the product of eigenvalues of A then yields the result

$$1 + \delta = \frac{1}{(1 - c(t)\lambda_1)(1 - c(t)\lambda_2)(1 - c(t)\lambda_3)}.$$

Commentary: This formula yields a lot of fun insights. For example, if $\lambda_i > 0$ then the DM distribution is collapsing in the direction of the ith eigenvector. You can imagine a blob of matter collapsing along one direction looking like a pancake. This is why some funny people called this "Zeldovich Pancake Theory" back in the day. Once $c(t) = 1/\lambda_i$, the perturbation has collapsed along the corresponding direction and you formed a sheet. If you've collapsed along two direction (e.g. $c(t) > 1/\lambda_1, 1/\lambda_2$) then you've formed a filament, and if you've collapsed along three direction that's a node. From this you can see how the cosmic web emerged pretty naturally in this approximation.

From here on out we will assume that $c\lambda_i \ll 1$ such that we are in regime where linear perturbation theory can be applied

(c) [2pt] First lets figure out the direction these particles will be flying in. Show in linear perturbation theory that $\delta = c(\nabla \cdot \boldsymbol{f})$. Suppose there exists some time t_0 where $c(t_0) = 1$ (you'll see this is possible in the next part) and let $\delta(t_0) = \delta_0$. Show that this implies

$$\boldsymbol{f}(\mathbf{k}) = -i\frac{\mathbf{k}}{k^2}\delta_0(\mathbf{k}).$$

This should make intuitive sense: the direction a particle moves in is set by the initial perturbations at the particle's initial positions.

Solution: The results simply follows from $\mathrm{Tr}(A)$ being the sum of the eigenvalues

of A. Expanding the result of part (b) to first order yields

$$1+\delta \approx 1+c(t)\Sigma_i\lambda_i = 1+c(t)\mathrm{Tr}\left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{q}}\right) = 1+c(t)\nabla\cdot\boldsymbol{f} \Rightarrow \delta = c(\nabla\cdot\boldsymbol{f}).$$

We can use Eq.(1) to replace δ with $\nabla^2\phi$ giving us

$$\frac{a\nabla\cdot(\nabla\phi)}{4\pi G\rho_{m,0}} = c(\nabla\cdot\boldsymbol{f}).$$

Choosing normalizations such that $c(t_0) = a(t_0) = 1$ yields

$$\boldsymbol{f} = \frac{\nabla\phi(t_0)}{4\pi G\rho_{m,0}}.$$

The Poisson equation in Fourier space yields $-k^2\phi_0 = 4\pi G\rho_{m,0}\delta_0$. Rewriting the above equation in Fourier space then gives us

$$\boldsymbol{f}(\mathbf{k}) = -i\frac{\mathbf{k}}{k^2}\delta_0(\mathbf{k}).$$

(d) [3pt] By combining Eq.(1) with (2), show to linear order that $c(t)$ is governed by the differential equation

$$\frac{d^2c}{da^2} + \frac{3}{2a}\frac{dc}{da} = \frac{3}{2a^2}c \Rightarrow c \sim a + (\text{decaying mode}).$$

We can neglect the decaying mode.
Hint: Compute $d^2\boldsymbol{x}/da^2$ to relate $(c, \boldsymbol{f})$ with $\nabla\phi$. Then take the gradient of everything (e.g. $\nabla\cdot d^2\boldsymbol{x}/da^2$). There should be a common $(\nabla\cdot\boldsymbol{f})$ that cancels from both sides.

Solution: In EdS we have that $H^2/H_0^2 = \Omega_{m,0}a^{-3} \Rightarrow H = H_0a^{-3/2}$. We can write

$$\frac{d\boldsymbol{x}}{da} = \frac{1}{\dot{a}}\frac{d\boldsymbol{x}}{dt} = \frac{p}{a^3H} \Rightarrow \frac{d^2\boldsymbol{x}}{da^2} = -\frac{3}{2a}\frac{d\boldsymbol{x}}{da} - \frac{\nabla\phi}{aH_0^2}.$$

Since $d\boldsymbol{x}/da = -\boldsymbol{f}dc/da$ and $d^2\boldsymbol{x}/da^2 = -\boldsymbol{f}d^2c/da^2$ we get

$$-\boldsymbol{f}\left[\frac{d^2c}{da^2} + \frac{3}{2a}\frac{dc}{da}\right] = -\frac{\nabla\phi}{aH_0^2}.$$

Taking the gradient of both sides gives us a $\nabla^2\phi \sim \delta \sim c(\nabla\cdot\boldsymbol{f})$ on the RHS. We can cancel out the common $\nabla\cdot\boldsymbol{f}$ and use $H_0^2 = 8\pi G\rho_{m,0}/3$ to then get the result

$$\frac{d^2c}{da^2} + \frac{3}{2a}\frac{dc}{da} = \frac{3c}{2a^2}.$$

The solution to this differential equation contains a growing mode that goes like $c \sim a$.

With this, we have derived the Zeldovich approximation in EdS:

$$\boxed{\boldsymbol{x} = \boldsymbol{q} + a\int\frac{d^3k}{(2\pi)^3}\frac{i\mathbf{k}}{k^2}e^{i\mathbf{k}\cdot\boldsymbol{q}}\delta_0(\mathbf{k}).} \tag{3}$$

(Bonus: Comparing Zeldovich to N-body simulation) [5pt]
I have uploaded to Canvas two files derived from an cosmological N-body simulation with 128^3 particles in a 512 Mpc/h box assuming $\Omega_{m,0} = 1$ (no Λ).

a. `HW07_delta_q_0.npy`: The initial perturbation field δ_0 in $\boldsymbol{q}$ space sampled on a uniform grid spanning the box.

b. `HW07_particle_pos.npy`: Position of all the particles at $z = 0$ after running the full N-body simulation with the initial conditions from (1)[1].

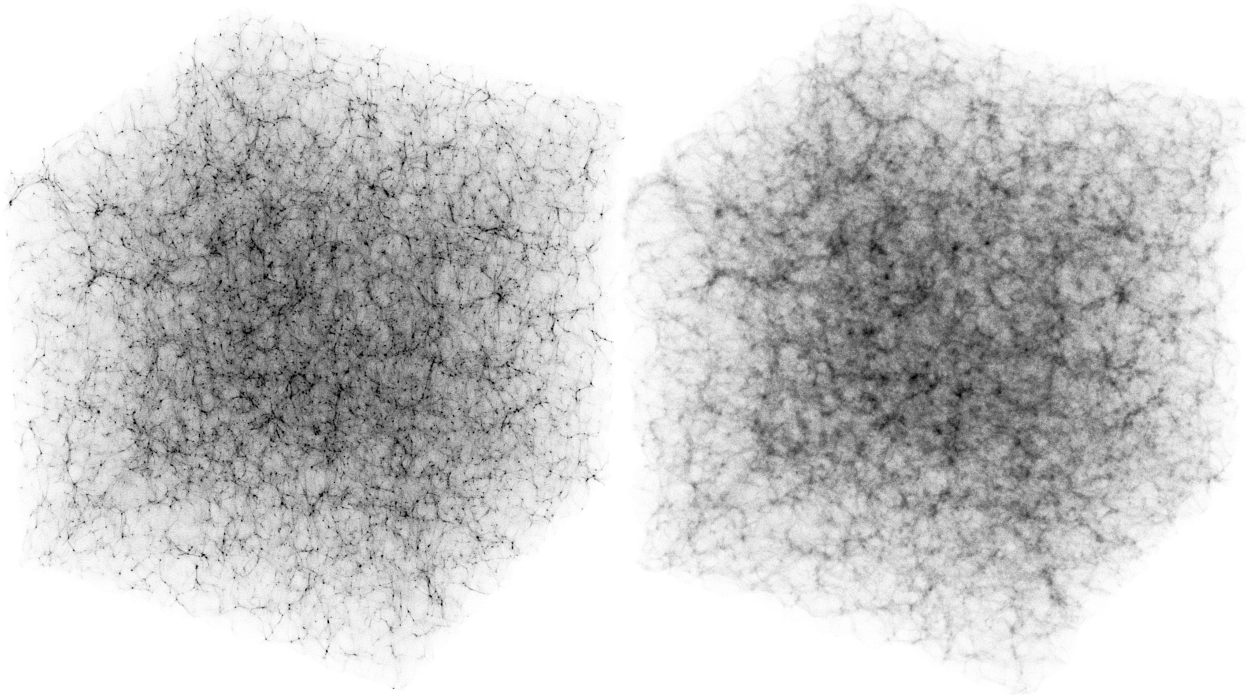
Both of these files can be loaded with `numpy.load`.

```
import numpy as np
delta_q_0 = np.load('HW07_delta_q_0.npy')
pos = np.load('HW07_particle_pos.npy')
```

[1] If you want positions at any other redshift, let me know. I have them.

Use Eq.(3) and `FFT`'s to make a perturbative prediction (Zeldovich) for how the cosmic web should look[2]. Compare this with the non-perturbative prediction from the N-body simulation. For this box, you should get something like this[3]

N-body Zeldovich



Solution: I uploaded a notebook `HW07_Bonus_Sol.ipynb` (and the HTML render `HW07_Bonus_Sol.html`) showing how I made the above two plots but alas didn't have time to make it pedagogical so it's a bit messy. There's actually a trick to getting much better "density" estimates by looking at the centroid of all the Lagrangian cells whose vertices are defined by the particles instead of the scatterplot of the particles themselves. If you'd like the details, see 1111.3944.

[2]Note: you must set $\boldsymbol{f}(\mathbf{k}=0)=0$ manually.

[3]There are other fun things to do. For example, you could try smoothing δ_0 and then looking at how the Zeldovich prediction changes. Or you could try making a movie of the cosmic web formation with just Eq.(3).